

Research Notes on Feature selection

Feature selection

>> Section 1

Here, $\overline{r_{cf}}$ is the average value of all feature-classification correlations, and $\overline{r_{ff}}$ is the average value of all feature-feature correlations. The CFS criterion is defined as follows:

>> Section 2

Application of feature selection metaheuristics This is a survey of the application of feature selection metaheuristics lately used in the literature. This survey was realized by J. Hammon in her 2013 thesis.

>> Section 3

Optimality criteria The choice of optimality criteria is difficult as there are multiple objectives in a feature selection task. Many common criteria incorporate a measure of accuracy, penalised by the number of features selected. Examples include Akaike information criterion (AIC) and Mallows's Cp, which have a penalty of 2 for each added feature. AIC is based on information theory, and is effectively derived via the maximum entropy principle. Other criteria are Bayesian information criterion (BIC), which uses a penalty of $\log \sqrt{n}$ for each added feature, minimum description length (MDL) which asymptotically uses $\log \sqrt{n}$, Bonferroni / RIC which use $2 \log \sqrt{p}$, maximum dependency feature selection, and a variety of new criteria that are motivated by false discovery rate (FDR), which use something close to $2 \log \sqrt{\frac{p}{q}}$. A maximum entropy rate criterion may also be used to select the most relevant subset of features.

>> Section 4

Quadratic programming feature selection mRMR is a typical example of an incremental greedy strategy for feature selection: once a feature has been selected, it cannot be deselected at a later stage. While mRMR could be optimized using floating search to reduce some features, it might also be reformulated as a global quadratic programming optimization problem as follows:

>> Section 5

Correlation feature selection The correlation feature selection (CFS) measure evaluates subsets of features on the basis of the following hypothesis: "Good feature subsets contain features highly correlated with the classification, yet uncorrelated to each other". The following equation gives the merit of a feature subset S consisting of k features:

>> Section 6

$$\text{SPEC}_{\text{CMI}} : \max_{\mathbf{x}} \{ \mathbf{x}^T \mathbf{Q} \mathbf{x} \} \text{ s.t. } \mathbf{x} = \mathbf{1}, \mathbf{x}_i \geq 0$$

>> Section 7

The r_{cf_i} and $r_{f_i f_j}$ variables are referred to as correlations, but are not necessarily Pearson's correlation coefficient or Spearman's ρ . Hall's dissertation uses neither of these, but uses three different measures of relatedness, minimum description length (MDL), symmetrical uncertainty, and relief. Let x_i be the set membership indicator function for feature f_i ; then the above can be rewritten as an optimization problem:

>> Section 8

Overview on metaheuristics methods A metaheuristic is a general description of an algorithm dedicated to solve difficult (typically NP-hard problem) optimization problems for which there is no classical solving methods. Generally, a metaheuristic is a stochastic algorithm tending to reach a global optimum. There are many metaheuristics, from a simple local search to a complex global search algorithm.

>> Section 9

Subset selection evaluates a subset of features as a group for suitability. Subset selection algorithms can be broken up into wrappers, filters, and embedded methods. Wrappers use a search algorithm to search through the space of possible features and evaluate each subset by running a model on the subset. Wrappers can be computationally expensive and have a risk of over fitting to the model. Filters are similar to wrappers in the search approach, but instead of evaluating against a model, a simpler filter is evaluated. Embedded techniques are embedded in, and specific to, a model. Many popular search approaches use greedy hill climbing, which iteratively evaluates a candidate subset of features, then modifies the subset and evaluates if the new subset is an improvement over the old. Evaluation of the subsets requires a scoring metric that grades a subset of features. Exhaustive search is generally impractical, so at some implementor (or operator) defined stopping point, the subset of features with the highest score discovered up to that point is selected as the satisfactory feature subset. The stopping criterion varies by algorithm; possible criteria include: a subset score exceeds a threshold, a program's maximum allowed run time has been surpassed, etc. Alternative search-based techniques are based on targeted projection pursuit which finds low-dimensional projections of the data that score highly: the features that

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have the largest projections in the lower-dimensional space are then selected. Search approaches include:

>> Section 10

$D(S, c) = \frac{1}{|S|} \sum_{f_i \in S} I(f_i; c)$ $\{\displaystyle D(S,c)=\frac{1}{|S|}\sum_{f_i \in S} I(f_i;c)\}$. The redundancy of all features in the set S is the average value of all mutual information values between the feature f_i and the feature f_j :

>> Section 11

where $F_{n \times 1} = [I(f_1; c), \dots, I(f_n; c)]^T$ $\{\displaystyle F_{n \times 1}=[I(f_1;c), \dots, I(f_n;c)]^T\}$ is the vector of feature relevancy assuming there are n features in total, $H_{n \times n} = [I(f_i; f_j)]_{i,j=1 \dots n}$ $\{\displaystyle H_{n \times n}=[I(f_i;f_j)]_{i,j=1 \dots n}\}$ is the matrix of feature pairwise redundancy, and $x_{n \times 1}$ $\{\displaystyle \mathbf{x}_{n \times 1}\}$ represents relative feature weights. QPFS is solved via quadratic programming. It is recently shown that QFPS is biased towards features with smaller entropy, due to its placement of the feature self redundancy term $I(f_i; f_i)$ $\{\displaystyle I(f_i;f_i)\}$ on the diagonal of H .

>> Section 12

$CFS = \max_k [r_{cf1} + r_{cf2} + \dots + r_{cfk} + 2(r_{f1f2} + r_{f1fj} + \dots + r_{fkfk} - 1)]$ $\{\displaystyle \mathrm{CFS} = \max_k [r_{cf1} + r_{cf2} + \dots + r_{cfk} + 2(r_{f1f2} + r_{f1fj} + \dots + r_{fkfk} - 1)]\}$. $\{\displaystyle \mathrm{CFS} = \max_k [r_{cf1} + r_{cf2} + \dots + r_{cfk} + 2(r_{f1f2} + r_{f1fj} + \dots + r_{fkfk} - 1)]\}$